

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B2: SYMMETRY AND RELATIVITY

TRINITY TERM 2015

Wednesday, 17 June, 2.30 pm – 4.30 pm

10 minutes reading time

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Larmor's formula for the power emitted by an accelerating point charge q is

$$P_L = \frac{\mu_0 q^2 a_0^2}{6\pi c} ,$$

where a_0 is the proper acceleration and the other symbols have their usual meaning.

1. A source moving through a laboratory with speed v emits a photon at angle θ to the direction of travel. Find a formula relating θ to the direction of emission in the rest frame of the source. Show that, for a source emitting photons isotropically in its rest frame, half the photons are emitted into a cone with half-angle $\cos^{-1}(v/c)$ in the laboratory.

[6]

Show that, for a body whose acceleration $\mathbf{a}$ is perpendicular to its velocity $\mathbf{v}$ in a given frame, the proper acceleration is $\gamma^2 a$, where $\gamma = (1 - v^2/c^2)^{-1/2}$.

[4]

A short bunch of electrons in a circular storage ring is used as a source of synchrotron radiation. The ring's radius is 561 m, and the energy of each electron is 3 GeV.

- (i) Find the radiative loss per revolution for each electron.
- (ii) Estimate the angular divergence of the emitted beam.
- (iii) Taking into account the fact that the source is moving, estimate the pulse-length for pulses received by a distant detector at a location fixed relative to the ring.
- (iv) Show that for a synchrotron of given radius, the spectrum of radiation received by a detector as in (iii) has a frequency spread proportional to γ^3 when the particle speed approaches the speed of light.

(v) In what part of the electromagnetic spectrum does the radiation mainly lie for the storage ring described?

[15]

2. Let f be a function defined over space and time, and let $df/d\tau$ refer to the rate of change of f , evaluated with respect to a worldline along which the elapsed proper time is τ . Show that

$$\frac{df}{d\tau} = u^\lambda \partial_\lambda f ,$$

where u^λ is the 4-velocity.

A manifestly covariant set of Euler-Lagrange equations may be written

$$\frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial u^a} \right) = \left(\frac{\partial \mathcal{L}}{\partial x^a} \right) ,$$

where x^a is the 4-position. In the case of motion of a point charge q of rest mass m in an electromagnetic field, the Lagrangian $\mathcal{L} = -mc(-u^\lambda u_\lambda)^{1/2} + qu^\lambda A_\lambda$, where A^a is the 4-vector potential. Derive the equation of particle motion

$$m \frac{du_a}{d\tau} = q F_{a\lambda} u^\lambda ,$$

and give an expression for F^{ab} in terms of A^a .

[8]

The spin 4-vector s^α has components $(\gamma \mathbf{s} \cdot \mathbf{v}/c, \gamma \mathbf{s})$ for a particle whose velocity is $\mathbf{v}$ and intrinsic angular momentum is $\mathbf{s}$ in a given frame, where $\gamma = (1 - \mathbf{v}^2/c^2)^{-1/2}$. In the absence of an applied torque, the spin 4-vector of an accelerating particle evolves as

$$\frac{ds^\alpha}{d\tau} = \frac{s_\lambda \dot{u}^\lambda}{c^2} u^\alpha ,$$

where the dot signifies $d/d\tau$.

(i) Show that the 4-spin remains orthogonal to the 4-velocity.

(ii) Show that the 3-spin $\mathbf{s}$ evolves as

$$\frac{d\mathbf{s}}{d\tau} = \frac{\gamma^2}{c^2} [(\mathbf{s} \cdot \dot{\mathbf{v}})\mathbf{v} - (\mathbf{v} \cdot \dot{\mathbf{v}})\mathbf{s}] .$$

(iii) Find $\mathbf{s}(\tau)$ for a particle accelerated along a straight line with speed $v(\tau) = c[1 - \exp(-2\Gamma\tau)]^{1/2}$, where Γ is constant.

[11]

The spin 4-vector of a charged particle of rest mass m and charge q moving in an electromagnetic field F evolves as

$$\frac{ds^a}{d\tau} = \frac{gq}{2m} \left(F^{a\lambda} s_\lambda + \frac{1}{c^2} F^{\mu\nu} u_\mu s_\nu u^a \right) + \frac{s_\lambda \dot{u}^\lambda}{c^2} u^a ,$$

where g is gyromagnetic ratio. Show that, if the particle's acceleration is due to the electromagnetic force (the Lorentz force), then the magnitude of the proper spin remains constant.

[6]

3. A high-energy plasma contains a large number of particles. The i th particle has energy E_i and 3-momentum $\mathbf{p}_i$.

(a) Under what conditions is the quantity $\sum_i (E_i^2 - p_i^2 c^2)$, summed over half of the particles, Lorentz-invariant?

(b) Under what conditions is the quantity $(\sum_i E_i)^2 - (\sum_i \mathbf{p}_i)^2 c^2$, summed over half of the particles, Lorentz-invariant? [4]

A neutral Kaon of rest mass M has energy E in the laboratory. It decays into a pair of charged pions, such that the first pion travels, in the laboratory, in a direction perpendicular to the initial velocity of the Kaon. Show that the momenta p_1, p_2 of the pions are related to that of the Kaon, p , by

$$p^2 = p_2^2 - p_1^2 .$$

Show that the energy of the second pion, in the laboratory, is given by

$$E_2 = E - \frac{M^2 c^4}{2E} .$$

Hence find the direction of travel of the second pion, and the momenta of both pions, when $E = 600$ MeV. [11]

It happens that the first pion survives for a proper time of 15 ns, and the second for a proper time of 30 ns. Find the distance and time between the pion decay events in the laboratory. What is the distance between the decay events in the frame in which they are simultaneous? [10]

[The rest mass energies are: K^0 : 497.6 MeV, π^+ : 139.6 MeV.]

4. A pair of powerful rockets accelerate continuously along a line, such that the distance between them remains constant when observed in a fixed inertial frame S. Without detailed calculation, describe qualitatively what happens to a string extended between the rockets and attached to both. [3]

Next suppose that the rockets move in such a way that the proper distance L between them remains constant. They move along the x axis of S, with rocket H always at larger x than rocket F. Let the trajectory of rocket F, relative to S, be given by $(x, y, z) = (f(t), 0, 0)$, where f is a known function. The trajectory of rocket H is then $(h(t), 0, 0)$, where the function $h(t)$ is to be discovered.

(i) Show that an event B at rocket H that is simultaneous with a given event A at rocket F, in the instantaneous rest frame of the latter, satisfies

$$c^2(t_B - t_A) = v [h(t_B) - f(t_A)] ,$$

where $v = df/dt$ evaluated at t_A and all the distances and times are relative to S.

(ii) Write down, in terms of h , f , t_B , t_A and L , the condition that the proper distance between the rockets remains equal to L .

(iii) Hence show that h is given by

$$h(t + \gamma Lv/c^2) = f(t) + \gamma L ,$$

making clear in your answer at what event the quantities v and γ are to be evaluated (in this formula). [9]

For the case $f(t) = (a^2 + c^2 t^2)^{1/2}$, where a is constant, find a simple expression for γv in terms of t , and hence, or otherwise, find an explicit expression for $h(t)$.

At $t = 0$ a pulse of light is emitted from rocket H and travels towards rocket F. Find the time and place of arrival of the pulse, and the speed of F, relative to S, when the pulse arrives at F. Show that the relationship between the frequencies ω of such a pulse, as observed in the two rockets, is

$$\frac{\omega_F}{\omega_H} = \frac{a + L}{a} ,$$

where it is assumed that each local observer agrees with what is observed by an inertial observer at rest in the instantaneous rest frame (you may assume the formula for the Doppler effect). [13]